# Data Engineering Assessment

Koro Data Team

## Instructions

Thank you for expressing interest in the Data Engineering role at Koro. As part of our hiring process, we would like you to complete the following task before your scheduled interview. Please take the time to carefully address the questions and provide detailed documentation of your approach. Submit your response as a single document, including any necessary SQL scripts and pseudocode.

## Scenario

Your company's ERP system stores e-commerce data in the following tables:

- **Orders Table:**
  - `order_id (pk)`
  - `amount`
  - `currency`
  - `order_status`
- **Order_customer Table:**
  - `id (pk)`
  - `order_id (fk)`
  - `customer_id (fk)`
- **Order_shipment Table:**
  - `id (pk)`
  - `order_id (fk)`
  - `address_id (fk)`
  - `tracking_code`
  - `shipping_status`
- **Order_billing Table:**
  - `id (pk)`
  - `order_id (fk)`
  - `address_id (fk)`

    - payment_method
    - payment_status
- **Address Table:**
    - address_id (pk)
    - name
    - street
    - zip_code
    - city
    - country
- **Customer Table:**
    - customer_id (pk)
    - name
    - company

The goal is to design a robust BigQuery data warehouse and implement a pipeline that ensures the accuracy, reliability, and usability of the data for analytics. The ERP system's API endpoint allows both push and pull API calls to access the data.

# Task 1: Schema Design and Implementation

**Design a schema that makes this data usable for analytics and create it using SQL, with a preference for the BigQuery flavor. Document everything, explaining your choices and highlighting the advantages and disadvantages of your chosen schema. Include examples of few common analytics queries. Also include discussions on considerations such as data correctness and anomalies**

Your schema should enable common analytics queries such as:

- Tracking revenue trends by country and currency.
- Analyzing customer activity and order history.
- Monitoring shipping performance and identifying delays.

.

—

# Task 2: Data Pipeline and Quality Assurance

**Design a data pipeline that extracts data from the ERP system's APIs and loads it into BigQuery while ensuring data accuracy and reliability. Document the pipeline design, addressing data quality checks and anomaly detection.**

Include details on:

- The GCP services you would choose to use.
- Strategies to validate data accuracy during the pipeline execution.
- Strategies to ensure data security considerations.

—

# Task 3: ELT with dbt and BigQuery Projects

**Scenario:** The BigQuery warehouse you designed has grown significantly, and there is now a need to restructure it. The team has decided to adopt an ELT (Extract, Load, Transform) approach using dbt (Data Build Tool). The pipeline you built earlier will remain in place, ingesting data into a `raw_data` project in BigQuery where access will be limited to engineers and analysts in the data team. You are tasked with researching the design of additional projects in BigQuery to support the following requirements:

- A `sandbox` project for running tests and experiments.
- A `production` project where final models are stored and accessed by analytics and visualization tools.
- Ensure other teams that do not need full access can access appropriate datasets or views in one of these new projects.

**Task:** - Discuss aspects of this design and discuss how you would structure datasets in these projects explaining how it addresses scalability and team-specific access needs. Document your design, Keep in mind the new datasets in the projects will be result of an ELT process that use the `raw_data` project tables. Explain how dbt fits into this structure and how it would transform raw data into usable models in the production project.

- Propose IAM roles and permissions to enforce security, ensuring:

- Engineers have full access to the `raw_data` project.
- Analysts and visualization tools can access only the `production` project.
- Developers can use the `sandbox` project for testing without impacting production data.

—

**Submission Instructions:** Please submit your documented responses along with any necessary scripts and documentation in a single document. We encourage you to attempt all questions to the best of your ability. If you find yourself stuck on a particular problem, move on to the next question and revisit the challenging one later. Your responses will be evaluated based on the following criteria:

1. **Schema Design and SQL Proficiency:**
   - Effectiveness of the schema for analytics.
   - Proper use of BigQuery-specific features.
2. **Pipeline Design and Robustness:**
   - Logical and scalable design of the data pipeline.
   - Ability to address edge cases and ensure reliable data ingestion.
3. **Data Quality and Monitoring:**
   - Thoughtful strategies for detecting and resolving anomalies.
   - Proactive monitoring and alerting mechanisms.
4. **ELT and Security with dbt:**
   - Clear design of BigQuery projects and datasets.

- Effective use of IAM roles to enforce access restrictions.
- Understanding of dbt and its role in the ELT process.

5. **Documentation and Communication:**
   - Clear explanations of design choices and trade-offs.
   - Well-structured and concise documentation.

Best of luck!

*Koro's Data Team*